Foundations of AlphaFold

[AlphaFold]

1.0 Content Block

Further reading Carlos Outeiral, CASP14: what Google DeepMind's AlphaFold 2 really achieved, and what it means for protein folding, biology and bioinformatics, Oxford Protein Informatics Group. (3 December) Mohammed AlQuraishi, AlphaFold2 @ CASP14: "It feels like one's child has left home." (blog), 8 December 2020 Mohammed AlQuraishi, The AlphaFold2 Method Paper: A Fount of Good Ideas (blog), 25 July 2021

2.0 Content Block

CASP14 In November 2020, DeepMind's new version, AlphaFold 2, won CASP14. Overall, AlphaFold 2 made the best prediction for 88 out of the 97 targets. On the competition's preferred global distance test (GDT) measure of accuracy, the program achieved a median score of 92.4 (out of 100), meaning that more than half of its predictions were scored at better than 92.4% for having their atoms in more-or-less the right place, a level of accuracy reported to be comparable to experimental techniques like X-ray crystallography. In 2018 AlphaFold 1 had only reached this level of accuracy in two of all of its predictions. 88% of predictions in the 2020 competition had a GDT_TS score of more than 80. On the group of targets classed as the most difficult, AlphaFold 2 achieved a median score of 87. Measured by the root-mean-square deviation (RMS-D) of the placement of the alpha-carbon atoms of the protein backbone chain, which tends to be dominated by the performance of the worst-fitted outliers, 88% of AlphaFold 2's predictions had an RMS deviation of less than 4 Å for the set of overlapped C-alpha atoms. 76% of predictions achieved better than 3 Å, and 46% had a C-alpha atom RMS accuracy better than 2 Å, with a median RMS deviation in its predictions of 2.1 Å for a set of overlapped CA atoms. AlphaFold 2 also achieved an accuracy in modelling surface side chains described as "really really extraordinary". To further validate AlphaFold 2, the conference organizers approached four leading experimental groups working on structures they found particularly challenging and had been unable to determine. In all four cases the three-dimensional models produced by AlphaFold 2 were sufficiently accurate to determine structures of these proteins by molecular replacement. These included target T1100 (Af1503), a small membrane protein studied by experimentalists for ten years. Of the three structures that AlphaFold 2 had the least success in predicting, one was an unusual multidomain complex consisting of 52 identical copies of the same domain. For all targets with a single domain, excluding only one very large protein and the two structures determined by NMR, AlphaFold 2 achieved a GDT_TS score of over 80.

3.0 Content Block

AlphaFold 1, 2, and AlphaFold DB AlphaFold DB provides models of individual protein chains (monomers), rather than their biologically relevant complexes. Many protein regions are predicted with low confidence score, including the intrinsically disordered protein regions. Alphafold-2 was validated for predicting effects of point mutations on structure and free energy, with a partial success.

4.0 Content Block

Unrelated AlphaFold-based databases isoform.io is a database of AlphaFold2-generated structures of proposed splice isoforms in the human genome. It includes information from 237,275 human transcripts. It has been used to detect errors in the mRNA predictions for a handful of genes.

5.0 Content Block

The AlphaFold Protein Structure Database (AFDB), a joint project between AlphaFold and EMBL-EBI, was launched on July 22, 2021. At launch, the database contained AlphaFold 1-predicted models for nearly the complete UniProt proteome of humans and 20 model organisms, totaling over 365,000 proteins. The database does not include proteins with fewer than 16 or more than 2700 amino acid residues, but for humans they are available in the whole batch file. AlphaFold's initial goal (as of early 2022) was to expand the database to cover most of the UniRef90 set, which contains over 100 million proteins. As of May 15, 2022, the database contained 992,316 predictions. In July 2021, UniProt-KB and InterPro has been updated to show AlphaFold predictions when available. On July 28, 2022, the team uploaded to the database the structures of around 200 million proteins from 1 million species, covering nearly every known protein on the planet. The number as of 2024 is 214 million, with 26 million being duplicates (exact sequence matches) of another protein in the database. The predicted structures can differ significantly between duplicates. As of 2025, the AFDB uses AlphaFold 2 for its predictions. All structures produced remain monomeric, but multimeric structures produced by other databases are linked on the page through the 3D-Beacons API. Foldseek, which provides fast and accurate structure searches, is also integrated. Information from AlphaMissense (a tool that uses AlphaFold to predict the outcome of missense mutations) is also integrated.

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Derived databases AlphaFill adds cofactors to AlphaFold models where appropriate. This is achieved by searching the Protein Data Bank for similar structures and transplanting cofactors to analogous positions. It is also linked to by UniProt. TmAlphaFold docks AlphaFold models to biological membranes, similar to what OPM does for PDB structures. AFTM uses AlphaFold models to identify transmembrane regions in human proteins, similar to what PDBTM does for PDB structures. ChannelsDB 2.0 uses PDB or AlphaFill models to calculate the pathway a molecule may take to reach an enzyme's active site or to reach another side of a transmembrane pore. The AFDB is not updated with UniProt sequences changes. AlphaSync keeps the AFDB in sync with UniProt entry changes, generating updated structures, residue-level features and contacts. It tries to use an AFDB entry for the exact updated sequence when available and run AlphaFold 2 independently otherwise. It fills in AFDB's blank for large (> 2,700 aa) proteins and proteins with special FASTA characters such as B, Z, U or X. The Encyclopedia of Domains (TED) applies the domain-recognition method from CATH database to 188 million unique structures from the AFDB, identifying nearly 365 million domains, which is 100 million more than what sequence-based methods could identify. The Evolutionary Classification of Protein Domains database (ECOD) assigns ECOD classifications to all SwissProt proteins in AFDB.

7.0 Content Block

Source code Open access to source code of several AlphaFold versions (excluding AlphaFold 3) has been provided by DeepMind in 2022 after requests from the scientific community. The source code and weights of AlphaFold 3 were made available for non-commercial use to the scientific community upon request in November 2024. It became publicly available in February 2025, still retaining the non-commercial restriction.